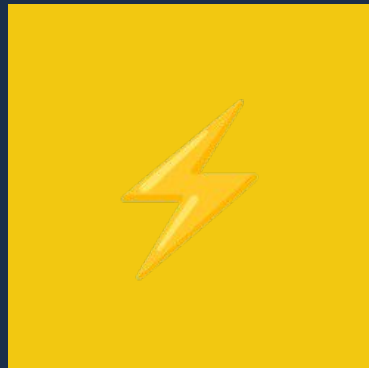


Power BI

Storage Modes

Explained simply — with real-world examples



What is a Storage Mode?

Think of it like a supermarket vs. a farmer's market

A supermarket stocks shelves in advance (fast, but may not have today's freshest). A farmer's market goes straight to the source every time (fresh, but takes longer).



WHERE is the data?

Stored inside Power BI or
left in the original source?



HOW FAST is it?

Query speed vs.
real-time freshness trade-off



HOW BIG can it be?

Dataset size limits
and memory use

1 / 4 — Import Mode

Data is **COPIED** into Power BI

- ✓ Fastest queries — data lives in memory
- ✓ Works offline, no live connection needed
- ✓ Supports all Power BI features (DAX, AI visuals)
- ⚠ Refreshed on schedule — NOT real-time
- ⚠ Dataset size limit: 1 GB (Pro) / 400 GB (Premium)

Real-world example



Monthly Sales Report

Finance team pulls last month's sales data from SQL — refreshes every morning at 6 AM.

No one needs it live minute-by-minute. Import is perfect here.

2 / 4 — DirectQuery Mode

Data STAYS at source — queried live

- ✓ Always up-to-date — real-time data
- ✓ No size limit — query as big as your DB
- ✓ Good for data that changes every minute
- ⚠ Slower — every click = new DB query
- ⚠ Limited DAX functions & transformations
- ⚠ Source DB must be always available

Real-world example



Live Operations Dashboard

Ops team monitors orders being placed RIGHT NOW. If data is 5 minutes old, decisions are wrong.

DirectQuery keeps the dashboard always current.

3 / 4 — Dual Mode & Composite Models

Dual Mode

Best of both worlds

The table can act as EITHER Import OR DirectQuery depending on what's being queried.

Analogy

Like a hotel restaurant: if you want a quick coffee, they serve from their stock (Import). If you want a special dish, they cook from fresh ingredients (DirectQuery).

Composite Model

Mix modes in one report

Some tables use Import (e.g. a product catalogue that rarely changes) and some use DirectQuery (e.g. live transactions).

Analogy

Like a food court: one stall sells pre-made sandwiches (Import = fast), another stall grills burgers to order (DirectQuery = fresh). You pick what you need from each.

Full Comparison — All Factors

Factor	Import	DirectQuery	Dual
Data Storage	Inside Power BI memory	Stays in source DB	Copy in PBI + can go live
Speed	Very fast	Slower (live queries)	Medium / optimized
Latency	Low	Higher	Medium
Memory Usage	High	Low	Moderate
Dataset Size	Small / medium	Huge datasets	Mixed scenarios
Real-Time Data	No (refresh-based)	Yes	Partial
Query Runs On	Power BI memory	Database / server	Mixed
DB Compute Cost	Lower	Higher	Moderate
Concurrent Users	Handles many well	Can overload DB	Better than DQ
DB Load	Low after refresh	High (always)	Medium
DAX Support	Full	Limited	Mostly supported
Transformations	Full Power Query	Some limitations	Partial
DB Dependency	Only during refresh	Always needed	Partial
Best For	Dashboards & reporting	Live monitoring	Enterprise hybrid
Main Advantage	Fastest performance	Always real-time	Balanced cost & speed
Main Limitation	Not real-time	Slow & expensive at scale	Complex setup